

Industrial Safety

Working at Heights Safety



Version 2.1 dated 1 Feb 2024

Working at Heights

Contents

1.	INTRODUCTION.....	3
2.	HEIGHT WORK.....	4
3.	HEIGHT WORK HAZARDS.....	8
4.	HEIGHT WORK ACCIDENTS	10
5.	RISKS AND IMPACTS.....	11
6.	SAFETY IN WORKING AT HEIGHTS	12
7.	SAFE PRACTICES	13
8.	PPE.....	15
9.	SUMMARY	16

Working at Heights

1. Introduction

Working at Height is where work is being done at a level higher than the floor level.

Usually, the work is done on stepladders, scaffolding, roof and even below ground levels.

These temporary platforms do not have permanent safeguards. The person involved can fall that can result in pains, bruises, broken bones or even death.



Working at Heights - Examples



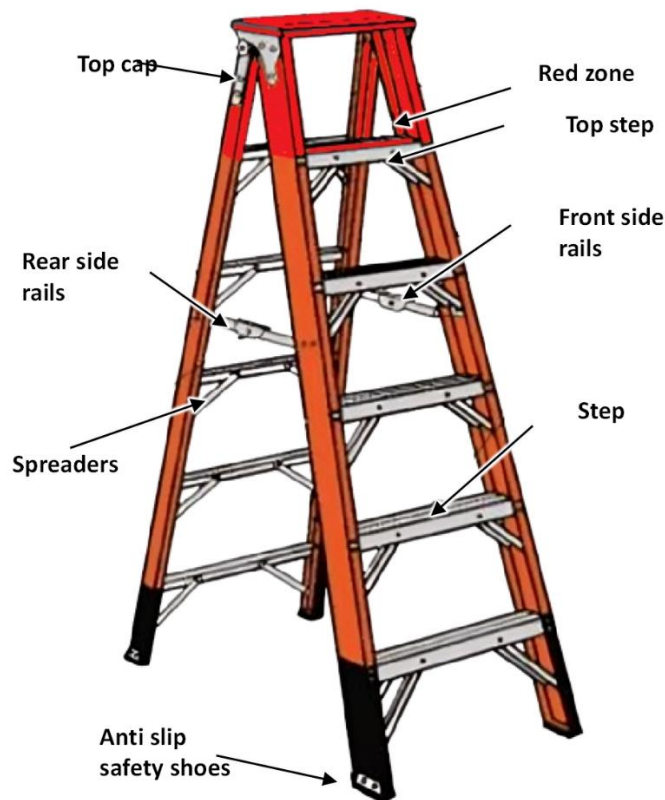
2. Height Work

A-Frame Ladder

Is commonly used for low height work. It is of two-legged ladder resembling an “A” shape, with steps on one or both sides. It is portable, foldable, provides stability on a flat surface and has a platform on top for keeping tools.

Safety rules

- ▶ Assign an assistant to hold ladder
- ▶ Ensure anti-slip shoes on ladder’s legs.
- ▶ Always practice 3-point contact on the ladder.
- ▶ Use lock pin to hold the opened ladder in place
- ▶ Don’t stand on Top-step or platform.



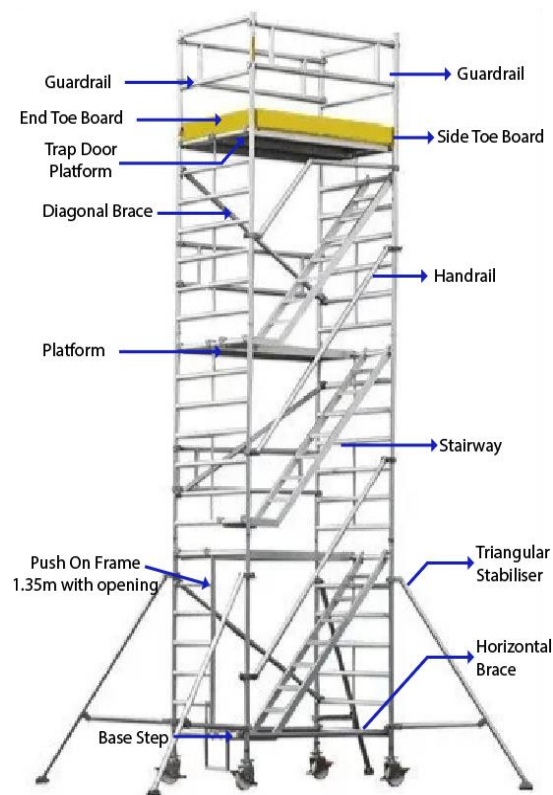
Working at Heights

Mobile Scaffold

Mobile scaffolds are set on wheels that can be moved around. They are commonly used for painting, plastering, installing and repair.

Scaffold Safety requirements:

- ▶ Structure should be of strong steel
- ▶ Fixed ladder to climb and descend
- ▶ Edge protection, Side rails, Side & cross bars
- ▶ Level footboards with Toe boards
- ▶ Wheels must be lockable
- ▶ Safety net to catch falling tools and objects
- ▶ Triangular anchor stabilizer at base
- ▶ No overloading (max weight on signboard)
- ▶ Sticker indicating inspection & fit for use



Working at Heights

Working on Roofs

Roof work means working on top of a building for the purpose of fixing the roof sheet, modifications, cleaning and maintenance, etc. If proper care is not taken, workers may fall down from the roof which may lead to fatal accidents.



Fragile Roof

A roof that does not have safety support and cannot withstand the weight of a person and any load the worker is carrying is termed as Fragile Roof.

Working at Heights

Some Facts

- ▶ Falls from heights during work are a major cause of serious injuries and deaths.
- ▶ Common cases of falls are from ladders and through unstable surfaces.
- ▶ Most injuries and fatalities happen from falls from four meters or less.
- ▶ Falls are the most common cause of head injuries, bone and hip fractures.
- ▶ Every year thousands of patients are hospitalized because of a fall injury, most often because of a head injury or hip fracture.



3. Height Work Hazards

Hazards in Height Work arise out of using substandard work platforms & unsafe acts of people.

Ladder, Bench, Stool

- ▶ Broken or loose rungs
- ▶ Floor instability
- ▶ Slippery walls

Scaffold, Sloping roof

- ▶ Platform instability
- ▶ Unguarded rails
- ▶ Missing platform planks
- ▶ Missing toe boards
- ▶ Overhead electric wires
- ▶ Screws & nails lying around

Others

- ▶ Untrained workers
- ▶ Improper PPE
- ▶ Improper behaviour

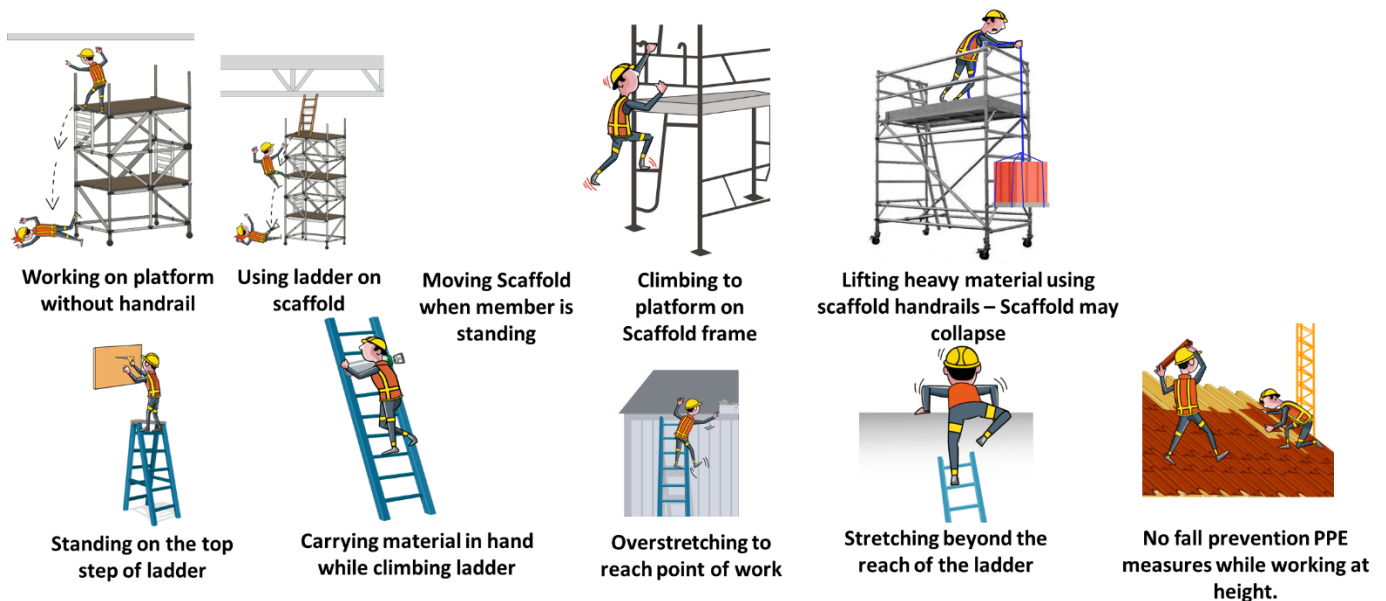


Working at Heights

Examples of Unsafe Conditions



Examples of Unsafe Acts



4. Height Work Accidents



5. Risks and Impacts

The impact of falling from heights could be:

- ▶ Pains
- ▶ Bruises
- ▶ Bone & hip fractures.
- ▶ Head injuries.
- ▶ Death



6. Safety in Working at Heights

Height Work involves high risk of Falls.

Falls can lead to long-term injuries or even death. Ways to prevent falls and reducing injury from fall impact is by

- ▶ Following Safe Practices
- ▶ Wearing Appropriate PPE, especially safety harness.



7. Safe Practices

Ladder, bench, stool

- ▶ Use a sturdy ladder
- ▶ Pre check ladder rungs and stability before use
- ▶ Use 3-point contact when climbing a ladder.
- ▶ Do not carry load while climbing up/down

Scaffold & Other Work platforms

- ▶ Inspect & certify scaffold before use and use sticker indicating fit for use
- ▶ Fix anchoring with Flat solid base plates and brakes on wheels
- ▶ Provide level footboards with hand rails and cross bars
- ▶ Provide Edge protection with Toe boards and side planks
- ▶ Safety net to catch falling tools and objects

Others:

- ▶ Take height work permit while working above 2 meters.
- ▶ Only trained people can work at the high place.
- ▶ Do not work at heights when feeling weak, sick, or dizzy.
- ▶ Display sign board for "Working on a High place".
- ▶ Do not throw material up or down or leave the materials & tools behind Do not work in poor weather like rainfall and strong winds.
- ▶ Wear a safety belt, fall arrestors and helmet while working at height.



Working at Heights

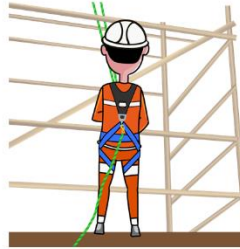
Safe Practices – Examples



Always maintain 3-Point contact while climbing ladder



Ensure ladder is held by supporting member



Always wear safety belt when performing height work



Use ladder to access scaffold



Ensure Hand rails, Guard rails and Toe boards are fixed in place.



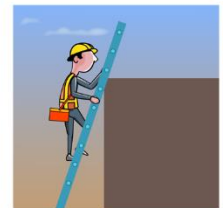
Use lifeline and full body harness when working on roof



Walk only of roof ladders



Walk only on roof crawling boards



Ladder should be projected 1 metre above the roof surface

8. PPE

PPE to be worn when working at heights include:

- ▶ Helmet
- ▶ Eye glasses
- ▶ High visibility jacket
- ▶ Body Harness with life-line
- ▶ Hand gloves
- ▶ Boots



PPE - Body Harness & Life Line

These are PPE to stop a worker from falling and prevent injury or death.



Use Safety harness, Lifeline and rope grab for safety when working at height.



Rope grab fall arrester, is connected between harness and vertical lifeline, will lock quickly and stop the fall of the worker.

Working at Heights

9. Summary

- ▶ Working at heights presents multiple hazards.
- ▶ Falls from heights during work are a major cause of serious injuries and deaths.
- ▶ Follow Safe Practices and wear PPE.
- ▶ Safety harness equipment play a big part in fall prevention.

